

Pharmacology of Drugs Used in Neurodegenerative Diseases: Alzheimer's disease (AD) and Parkinson's Disease (PD)



Pongsatorn Meesawatsom BPharm, PhD
Department of Pharmacology
Faculty of Pharmacy
Mahidol University



วัตถุประสงค์

★ Alzheimer's disease

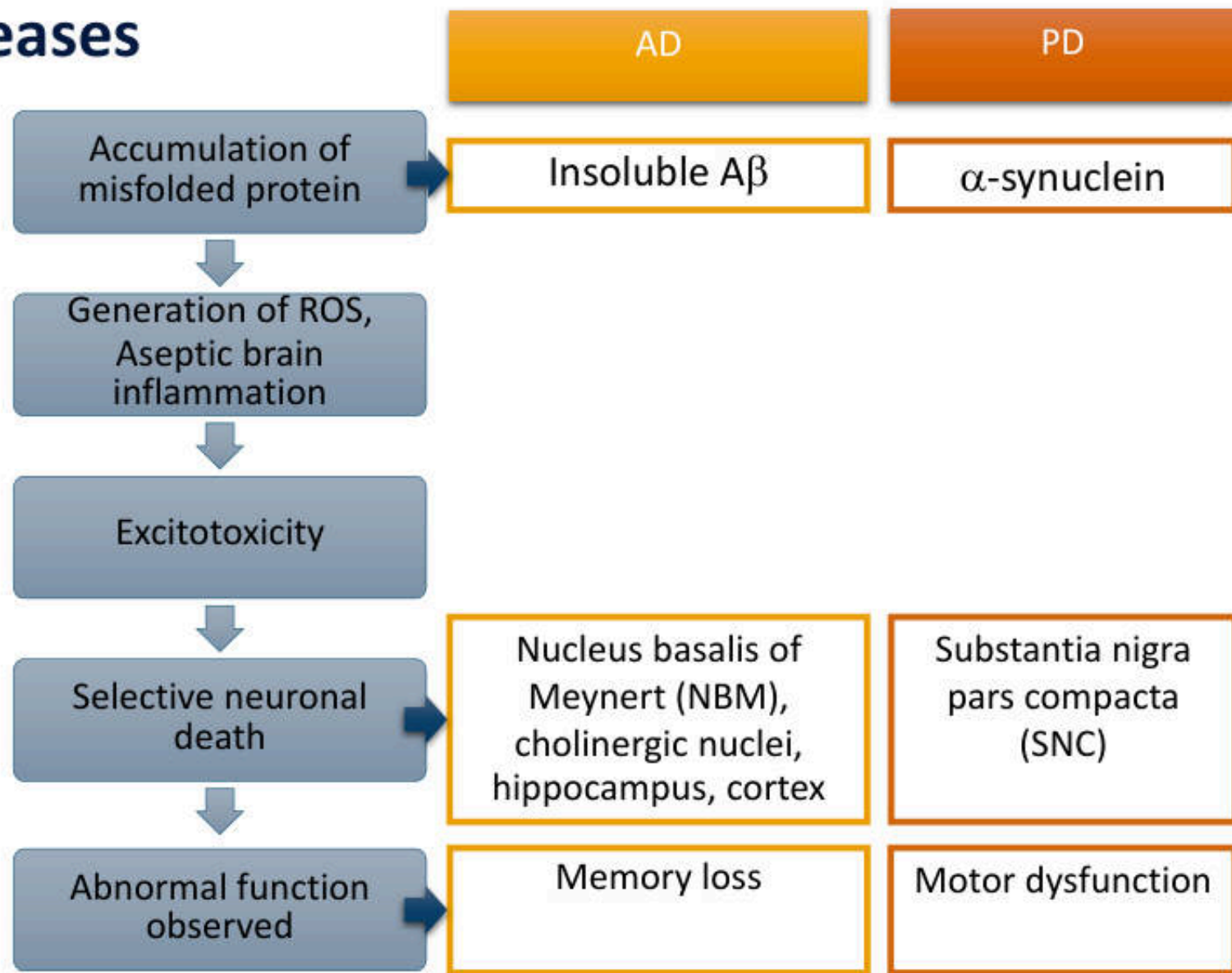
- ◇ อธิบายกลไกการออกฤทธิ์และผลทางเภสัชวิทยาของยาที่ใช้รักษา AD ได้
- ◇ เปรียบเทียบความแตกต่างทั้ง PK และ PD ที่สำคัญของ ChEIs ได้
- ◇ อธิบาย drug interaction ที่สำคัญของ ChEIs และ ผลทางคลินิกได้

★ Parkinson's Disease (PD)

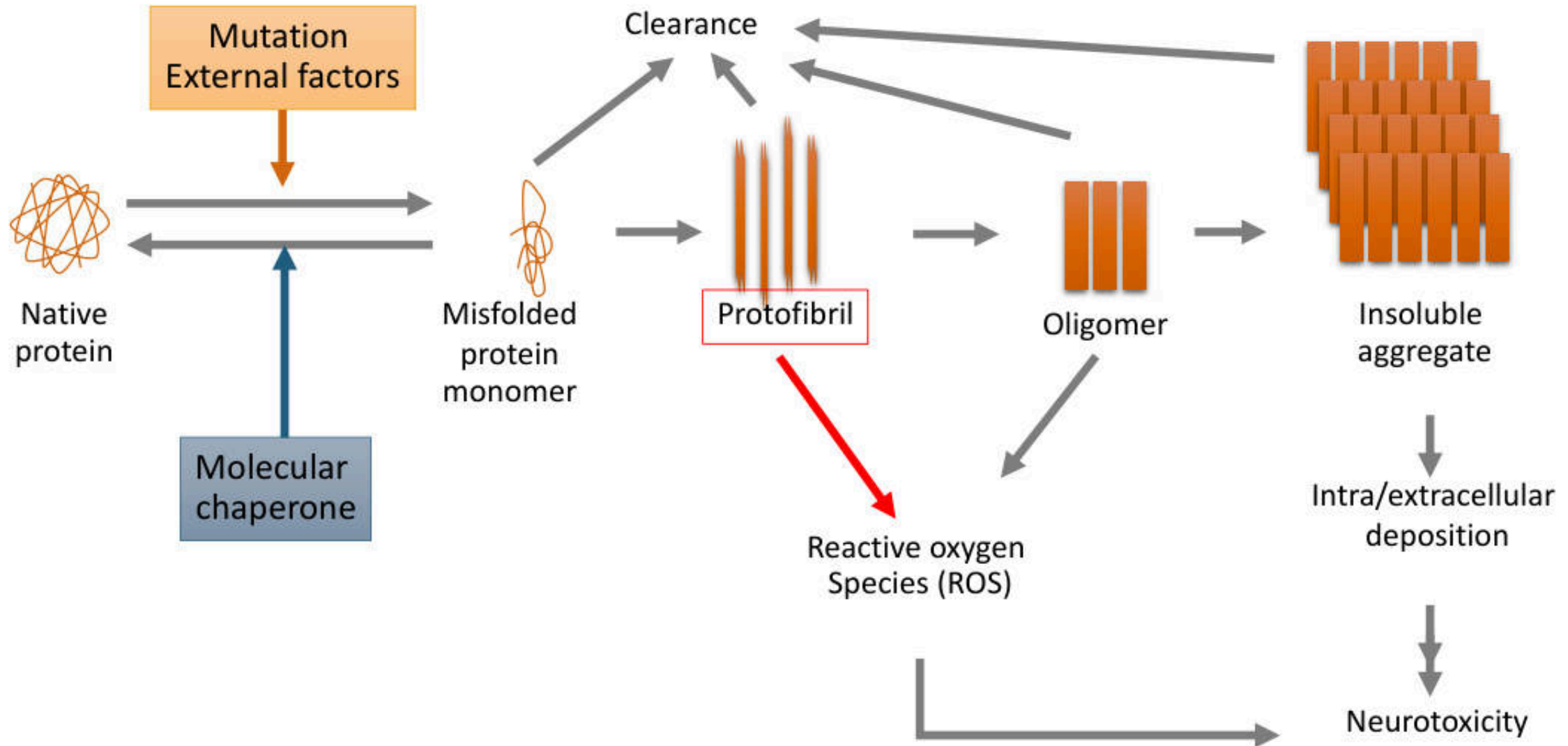
- ◇ อธิบายกลไกการออกฤทธิ์และผลทางเภสัชวิทยาของยาที่ใช้รักษา PD ได้
- ◇ อธิบายเภสัชวิทยาที่สำคัญของ levodopa ได้
 - ✦ PK
 - ✦ Combination
 - ✦ Motor fluctuation
- ◇ เปรียบเทียบผลการรักษา PD และอาการไม่พึงประสงค์ของยาแต่ละกลุ่มที่ใช้รักษา PD
- ◇ อธิบาย drug interaction ที่สำคัญของยาที่ใช้รักษา PD และ ผลทางคลินิกได้

Neurodegenerative diseases

- ★ Neurological disorders with gradual progression and most are likely irreversible
- ★ Age dependent
- ★ Neuronal loss occurs long before the onset of symptoms
- ★ No cure or treatment that stop the progression



Processes of Accumulation of misfolded proteins in neurodegenerative diseases

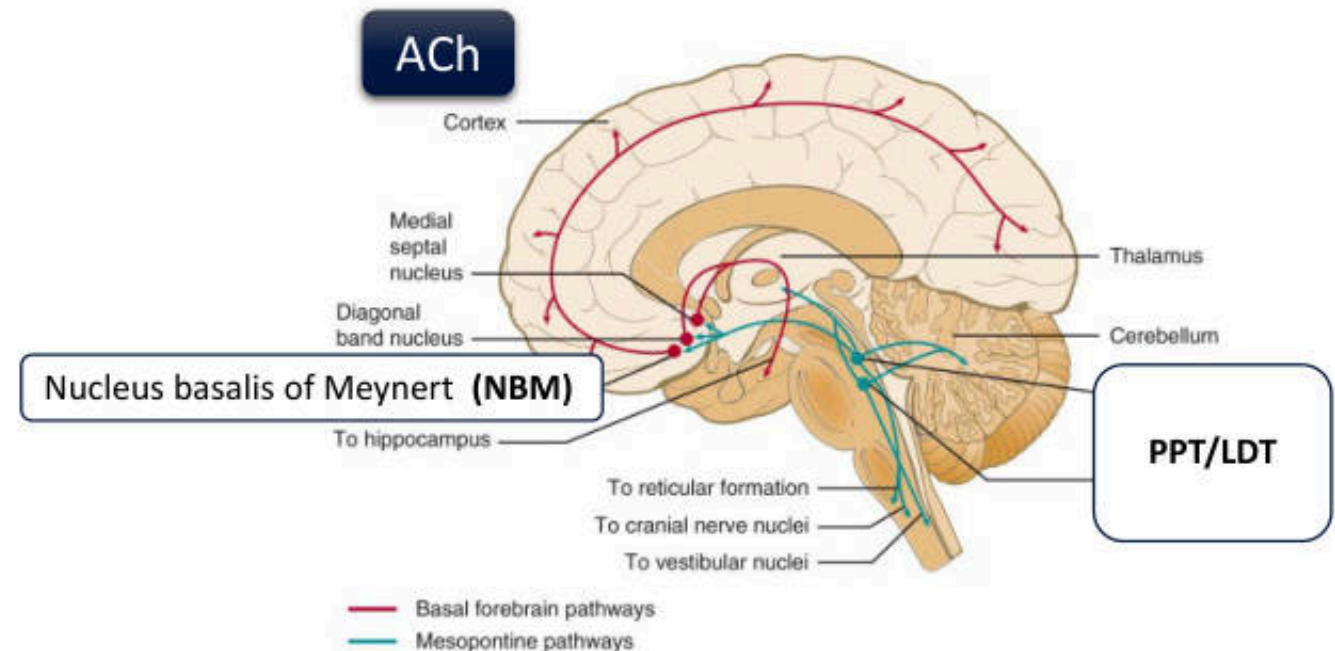
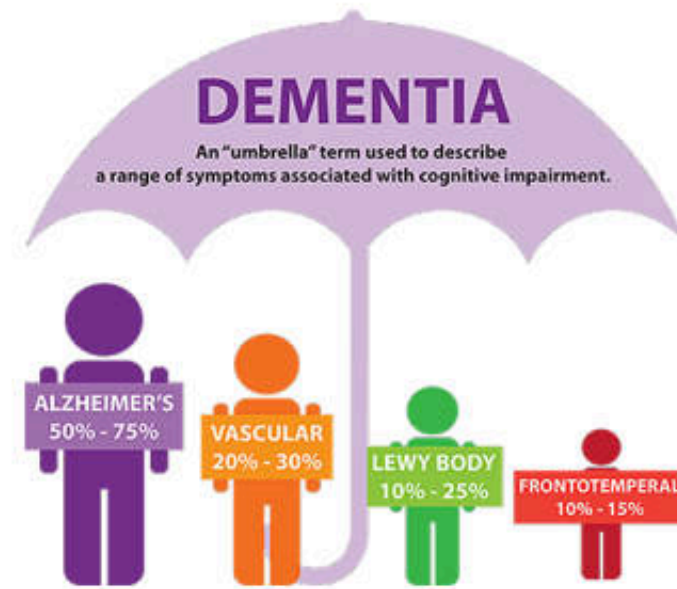


Drugs used in Alzheimer's Disease (AD)



Alzheimer's disease (AD)

- ★ Alzheimer's disease (AD) is the most common form of dementia
- ★ Age-related progressive neurodegenerative disorders
- ★ Results from **cholinergic pathway degeneration** lead to loss of memory & intellectual ability
 - ◆ 20% loss in mild to moderate symptom
 - ◆ 50% loss in moderate to severe



AD symptoms

★ Cognitive decline

- ◆ Aphasia, apraxia, agnosia and disturbance in executive functioning
- ◆ Impairment in activity of daily living (ADL)

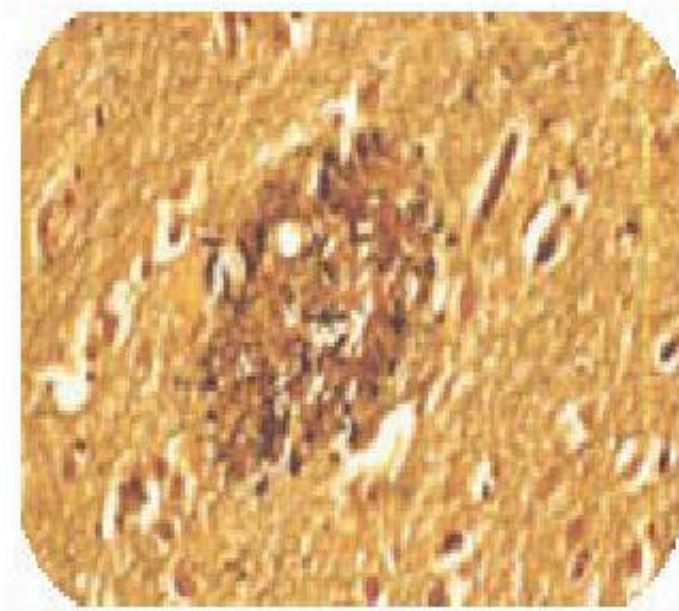
★ Behavioral and psychological symptoms of dementia (BPSD)

- ◆ Behavioral; disinhibition, aggression, wandering
- ◆ Psychological; depression, anxiety, insomnia, psychosis

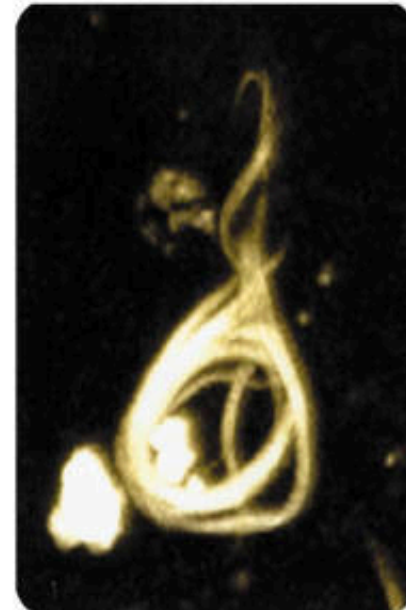
Important pathophysiology

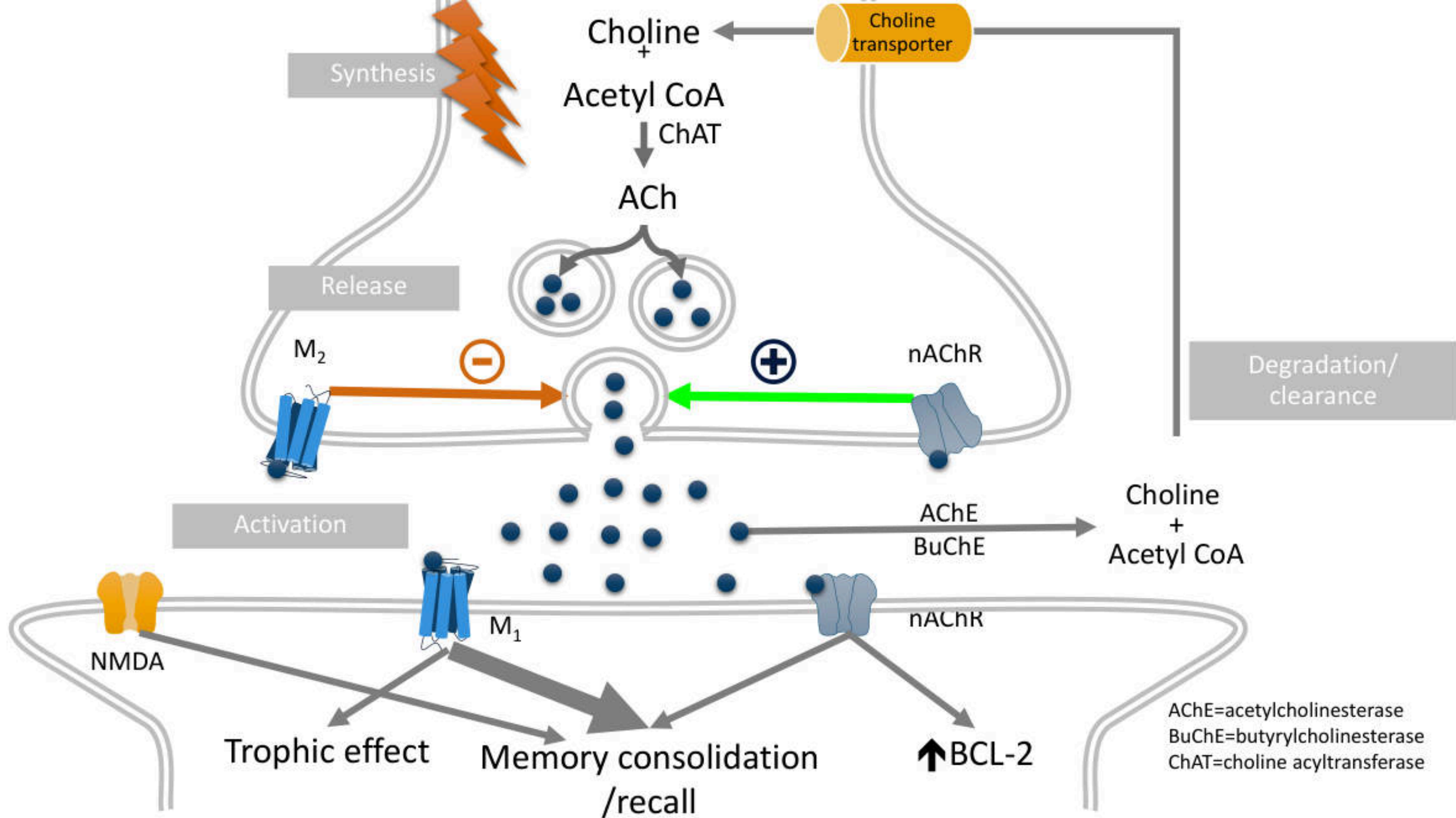
- ★ Extracellular deposition of amyloid protein aggregate → amyloid plaque
- ★ Intracellular deposition of hyperphosphorylated tau → neurofibrillary tangle

Amyloid plaque



Neurofibrillary tangle





Drug classes

- ★ Cholinesterase inhibitors (ChEIs)
 - ◆ First generation :physostigmine, tacrine
 - ◆ Second generation: for mild to moderate
 - ❖ Donepezil (Aricept[®], Aricept[®] Evess) for mild, moderate, severe dementia
 - ❖ Rivastigmine (Exelon[®], Exelon[®] patch) for mild, moderate, severe
 - ❖ Galantamine (Reminyl[®]) for mild to moderate
- ★ NMDA receptor antagonist: memantine (Ebixa[®])
- ★ *Ginkgo biloba* standardized extract: EGb761 (Tanakan[®])
- ★ Monoclonal antibodies against A β : aducanumab (Aduhelm[®]), lecanemab (Leqembi[®]), donanemab (Kisunla[®])

Pharmacodynamics of ChEIs

	Donepezil	Rivastigmine	Galantamine
AChE inhibition	✓	✓ Pseudoirreversible	✓
BuChE inhibition		✓	
α_7 nAChR APL			✓
σ receptor agonist	✓		

AChE=acetylcholinesterase

BuChE=butyrylcholinesterase

nAChR= nicotinic acetylcholine receptor

APL=allosteric potentiating ligand

Pharmacokinetics of ChEIs

Parameters	Donepezil	Rivastigmine	Galantamine
Tmax (hr)	3-5	0.5-2	0.5-1
Protein binding (%)	96	40	18
Plasma half-life (hr)	70-80	2	5-7
Metabolism	CYP2D6, CYP 3A4	Nonhepatic	CYP2D6, CYP 3A4
Dosing frequency	OD	BID	BID

ADRs of ChEI due to increasing cholinergic transmission

★ Periphery - > Cholinergic side effects

- ◇ Muscle cramp
- ◇ Nausea, vomiting, diarrhea, abdominal cramp
- ◇ Bradycardia, bronchoconstriction
- ◇ May increase risk of peptic ulcer

★ CNS

- ◇ Extrapyraxidal symptoms (mild)
- ◇ Sleep disturbance
- ◇ Weight loss
- ◇ Seizure

Pharmacodynamic drug interactions

★ Anticholinergic drugs

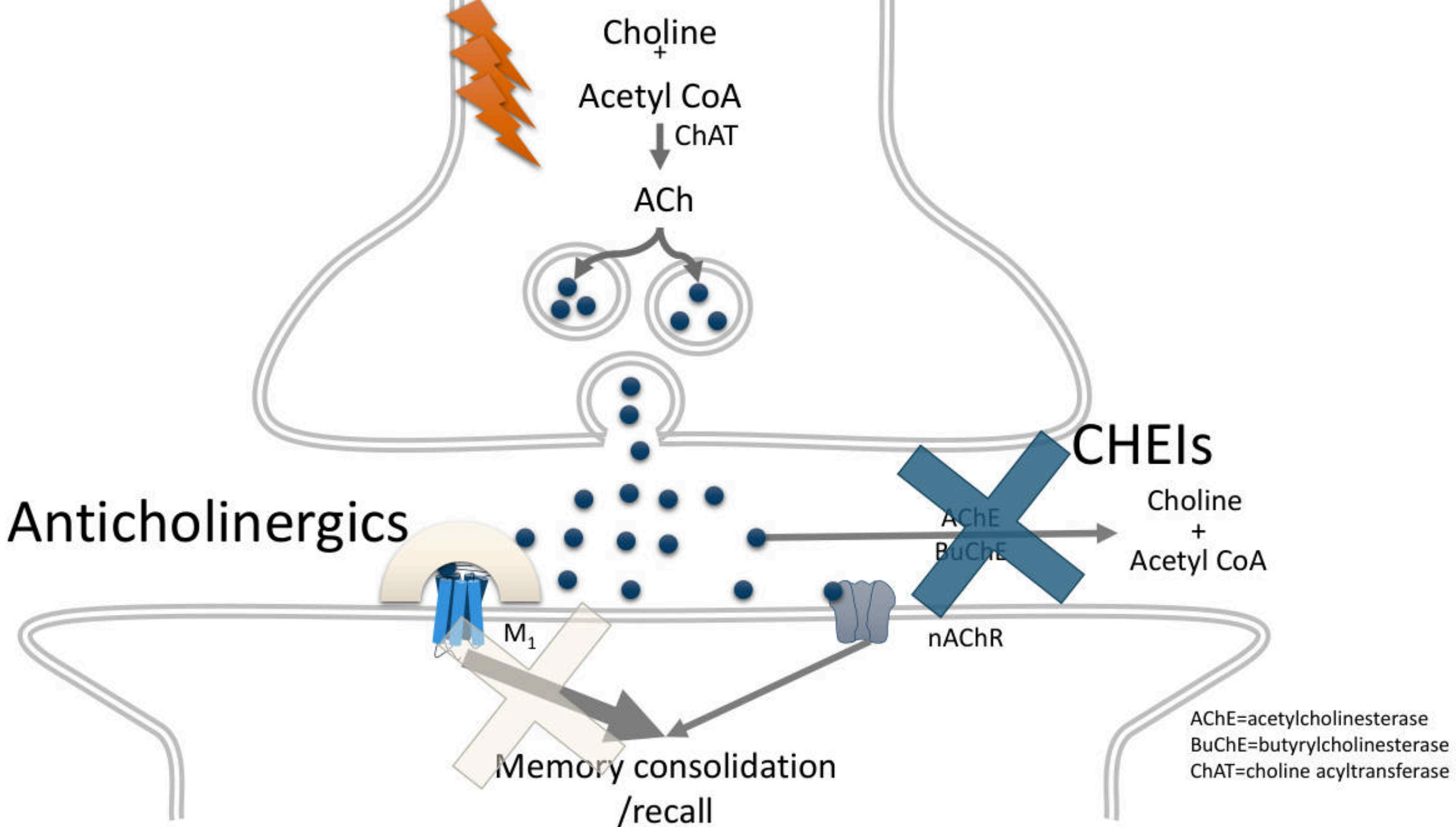
- ◇ Sedating antihistamines; chlorpheniramine, brompheniramine, triprolidine
- ◇ Hyoscine
- ◇ Benztropine, trihexyphenidyl
- ◇ Orphenadrine
- ◇ Tricyclic antidepressants
- ◇ Conventional antipsychotics

★ Cholinergic drugs

- ◇ Bethanechol

★ Antihypertensive drugs

- ◇ β -blockers
- ◇ Verapamil, diltiazem

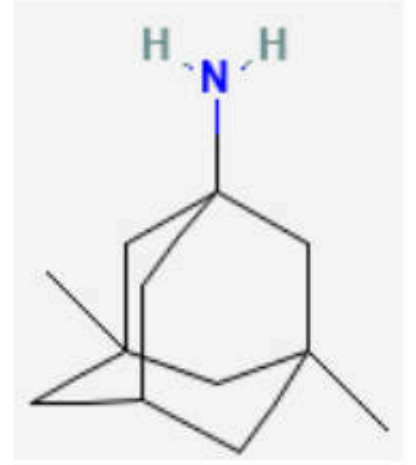


Pharmacokinetic drug interactions

- ★ Donepezil and galantamine are metabolized by CYP2D6 and CYP3A4
- ★ Enzyme inducer
 - ◇ CYP3A4; phenytoin, carbamazepine, rifampin
- ★ Enzyme inhibitor
 - ◇ CYP2D6; fluoxetine, paroxetine
 - ◇ CYP3A4; ketoconazole, clarithromycin

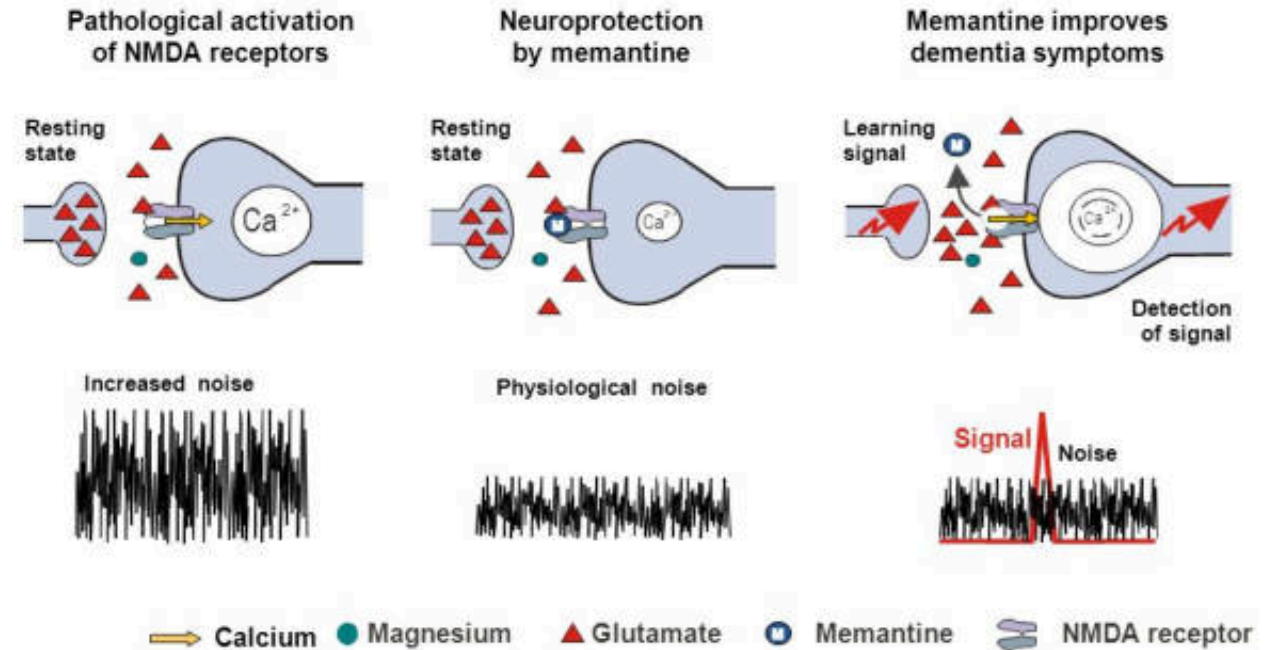
NMDA receptor antagonist: Memantine (Ebixa®)

- ★ Approved in moderate to severe AD
- ★ MOA: Partially block of NMDA receptor with moderate affinity
- ★ Improve long-term potentiation (LTP) of hippocampus
- ★ Excreted unchanged by kidney
- ★ Adverse effects
 - ◇ Most are not severe; dizziness, constipation, headache
 - ◇ Sedation, confusion, hallucination



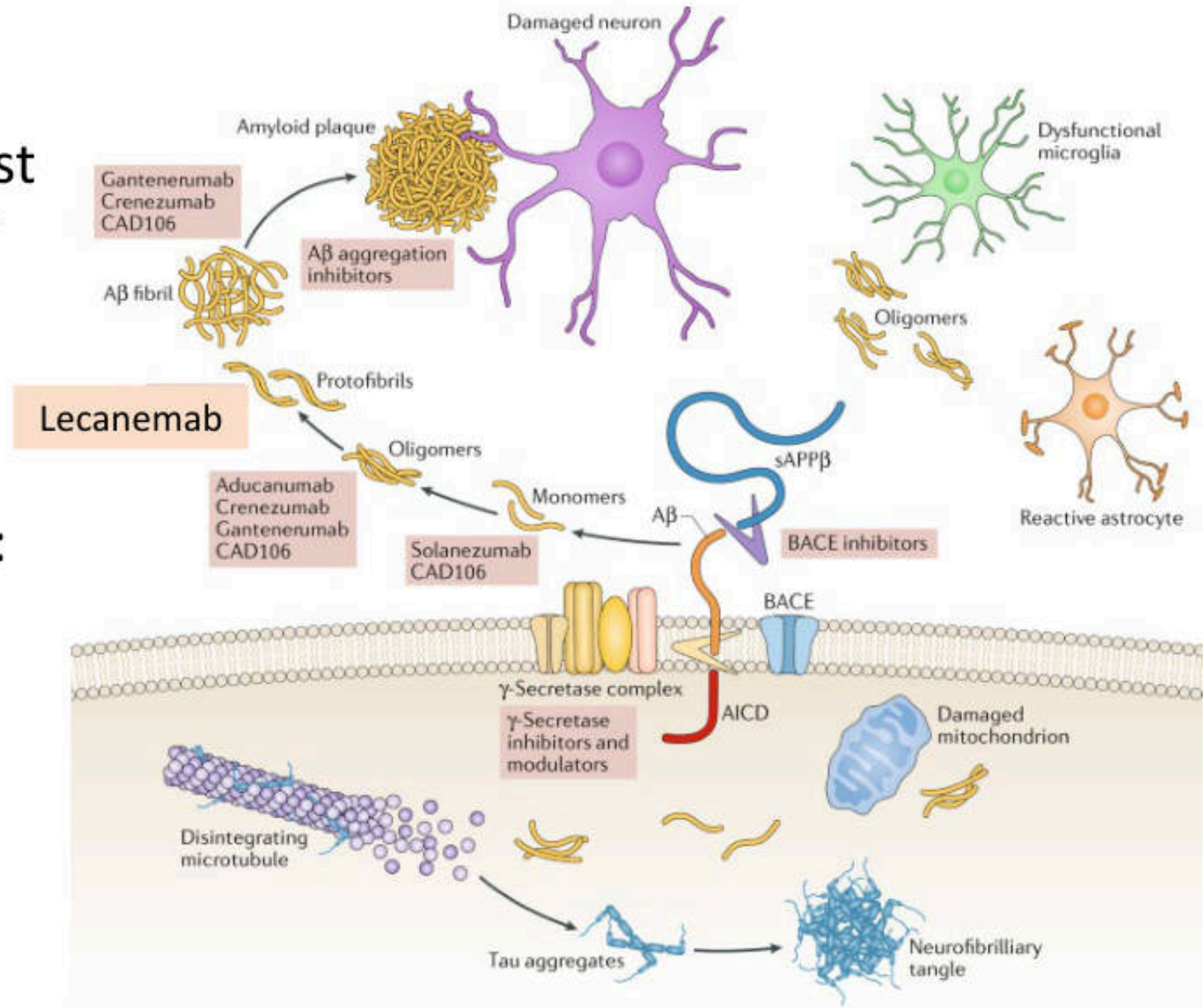
LTP improving mechanism by memantine

- ★ In moderate to severe AD, excess synaptic glutamate causes tonic overactivation of NMDA receptors which creates background noise and interfere LTP
- ★ Memantine block NMDA receptor and reduces background noise
- ★ However, memantine allows postsynaptic NMDA receptor activation during phasic release of glutamate and LTP induction



Novel targets for AD

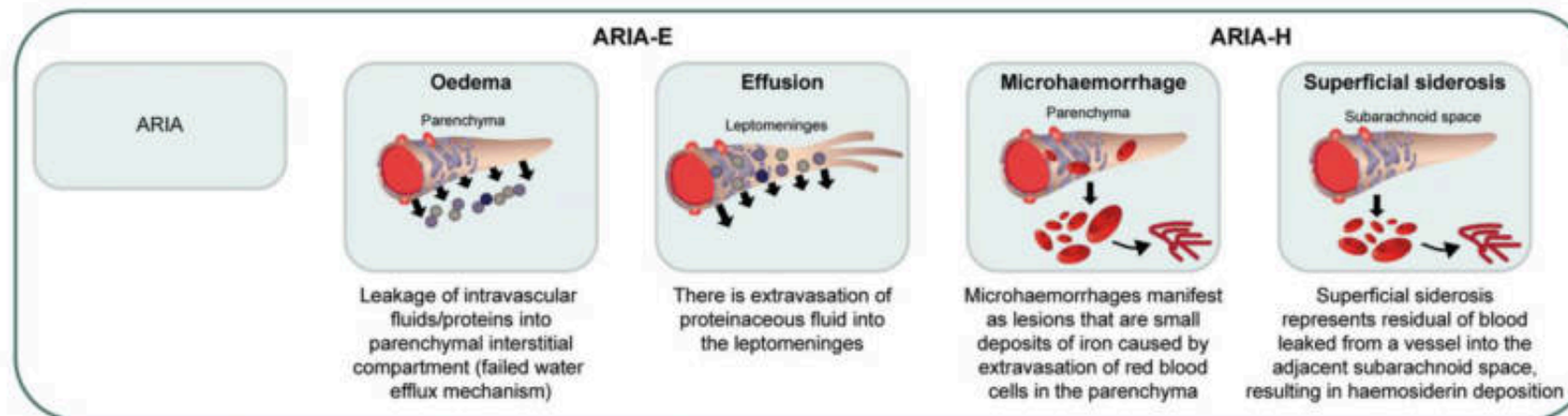
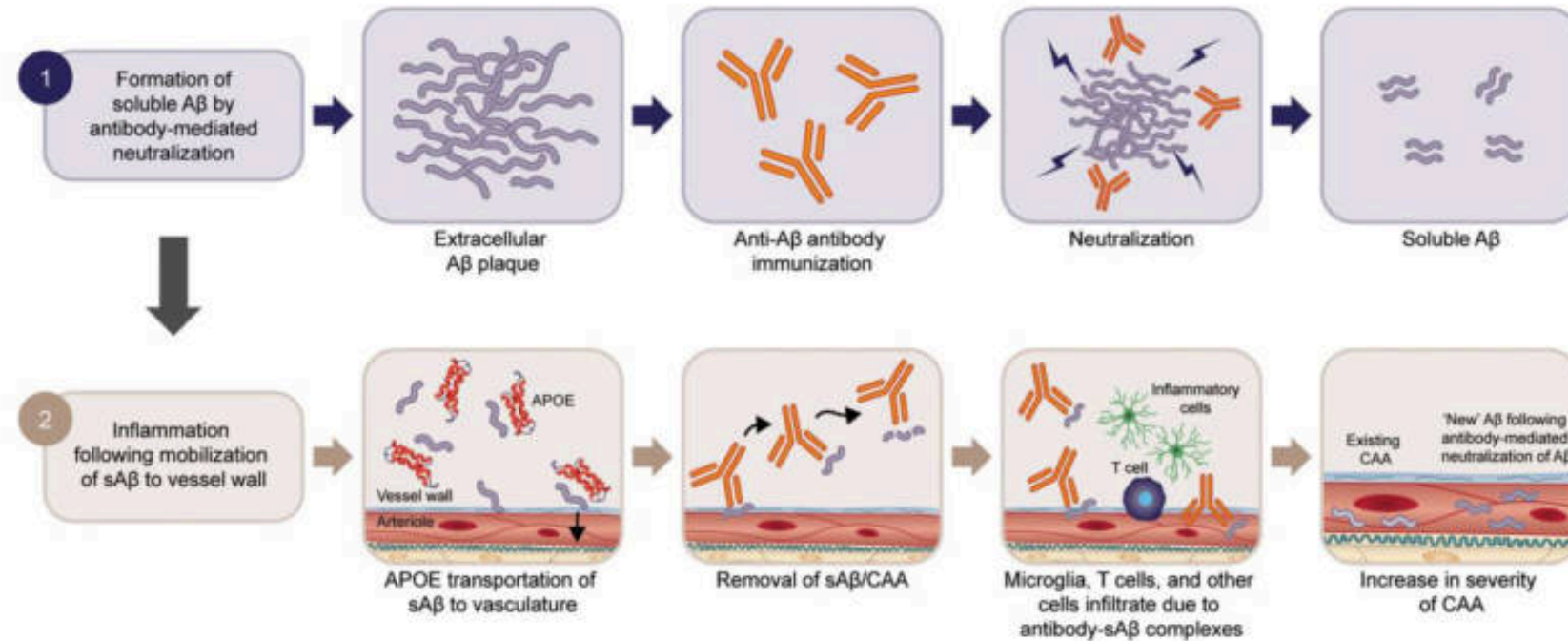
- ★ Monoclonal antibodies against amyloid peptide, oligomer or plaque: lecanemab, aducanumab, donanemab
- ★ β -secretase (BACE) inhibitors: verubecestat
- ★ γ -secretase inhibitors: semagacestat



Lecanemab (Leqembi®): A monoclonal antibody against amyloid-beta (A β) protofibrils

- ★ MOA: Bind to A β protofibrils – the most neurotoxin form of A β protein
- ★ Pharmacological effects
 - ◇ Reduces brain amyloid-beta plaques
 - ◇ Slows clinical and cognitive decline, roughly 30% over 18 months
- ★ Indication
 - ◇ For early AD – mild dementia or mild cognitive impairment
- ★ Conditions of use
 - ◇ Apolipoprotein E ϵ 4 (ApoE ϵ 4) non-carriers or heterozygotes
 - ◇ Confirmed amyloid pathology
- ★ Serious ADRs
 - ◇ Amyloid Related Imaging Abnormalities (ARIA) – monitored by MRI
 - ◇ ARIA-E - edema/swelling
 - ◇ ARIA-H - microhemorrhages/bleeding

Proposed pathophysiological mechanisms for ARIA



Drugs used in Parkinson's Disease (PD)



Parkinson's Disease (PD)

- ★ Progressive neurodegenerative disease with deposition of Levy body
- ★ 50-60% loss of dopamine producing neuron of SNC at onset of symptom
- ★ Current medications are symptomatic treatment, never cure PD but just slow the progression

Classical PD symptoms

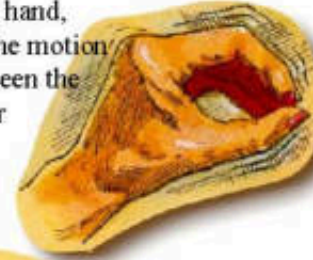
★ Motor

- ◆ Bradykinesia
- ◆ Resting tremor
- ◆ Rigidity
- ◆ Postural instability
- ◆ Others; micrographia, freezing, masked face, hypophonia, drooling

★ Non-motor

- ◆ Cognitive deficit
- ◆ Psychiatric; depression, anxiety, insomnia, psychosis
- ◆ Autonomic dysfunction
- ◆ Others

Rhythmic tremor often occurs at first in one hand, where it resembles the motion of rolling a pill between the thumb and forefinger



Leaning forward or backward when upright reflects impairment of balance and coordination.



Muscle rigidity shows itself in the cogwheel phenomenon: pushing on an arm causes it to move in jerky increments instead of smoothly.

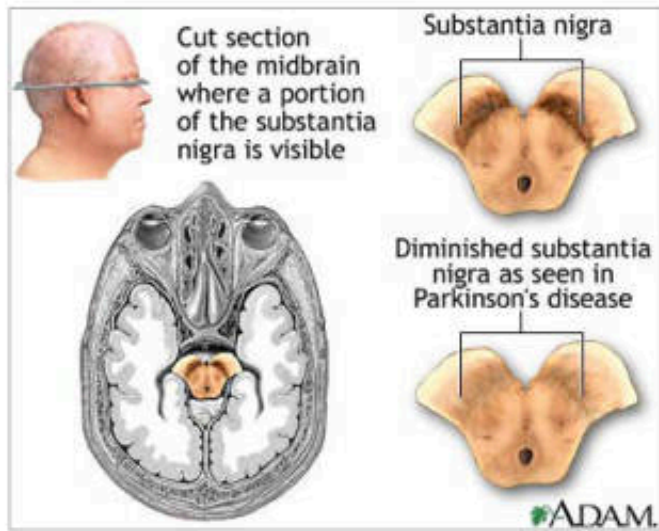


Difficulty rising from a sitting position is a common sign of disordered control over movement. Some patients report feelings of weakness and of being constrained by ropes or other forces.

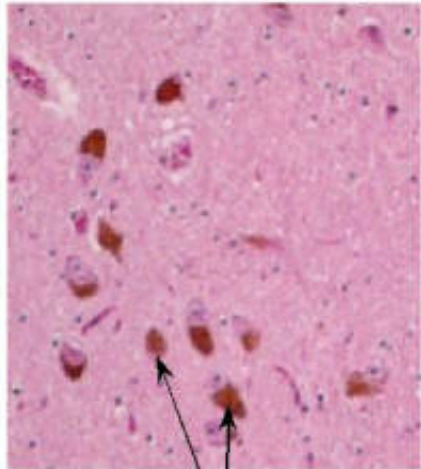
Parkinson's disease

I cannot control handwriting.

Today, over 4 million people in the US have Alzheimer disease. It is found mostly in people aged 65 or older.

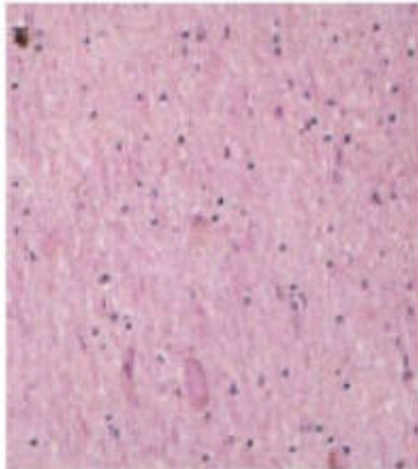


Normal Substantia Nigra

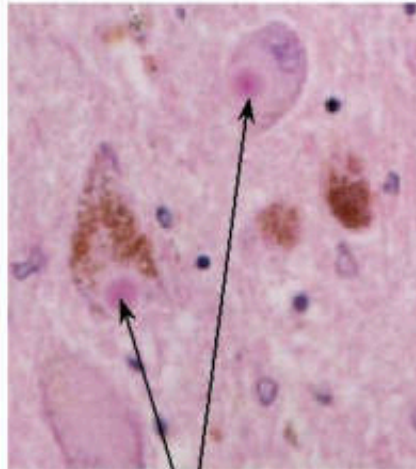


Pigmented neurons

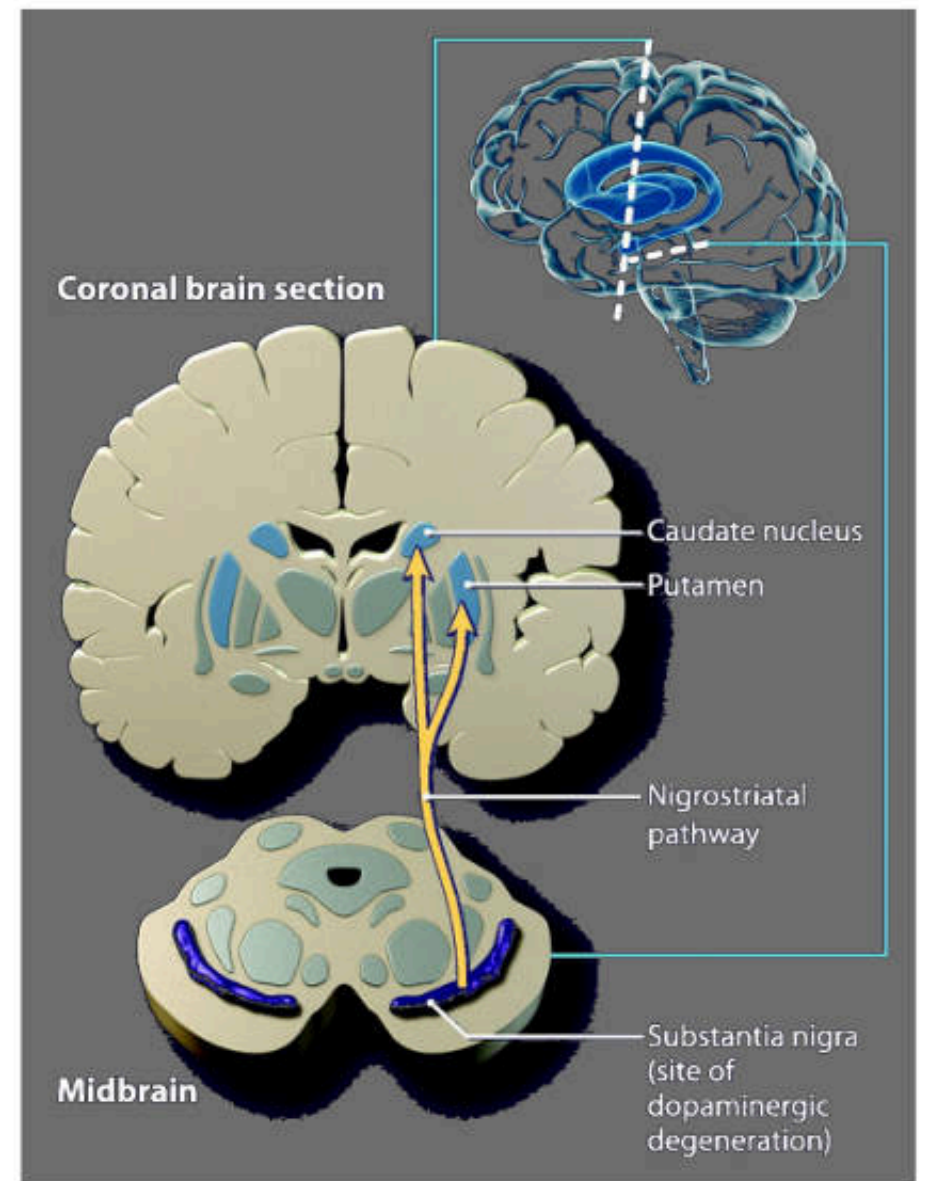
Parkinson's

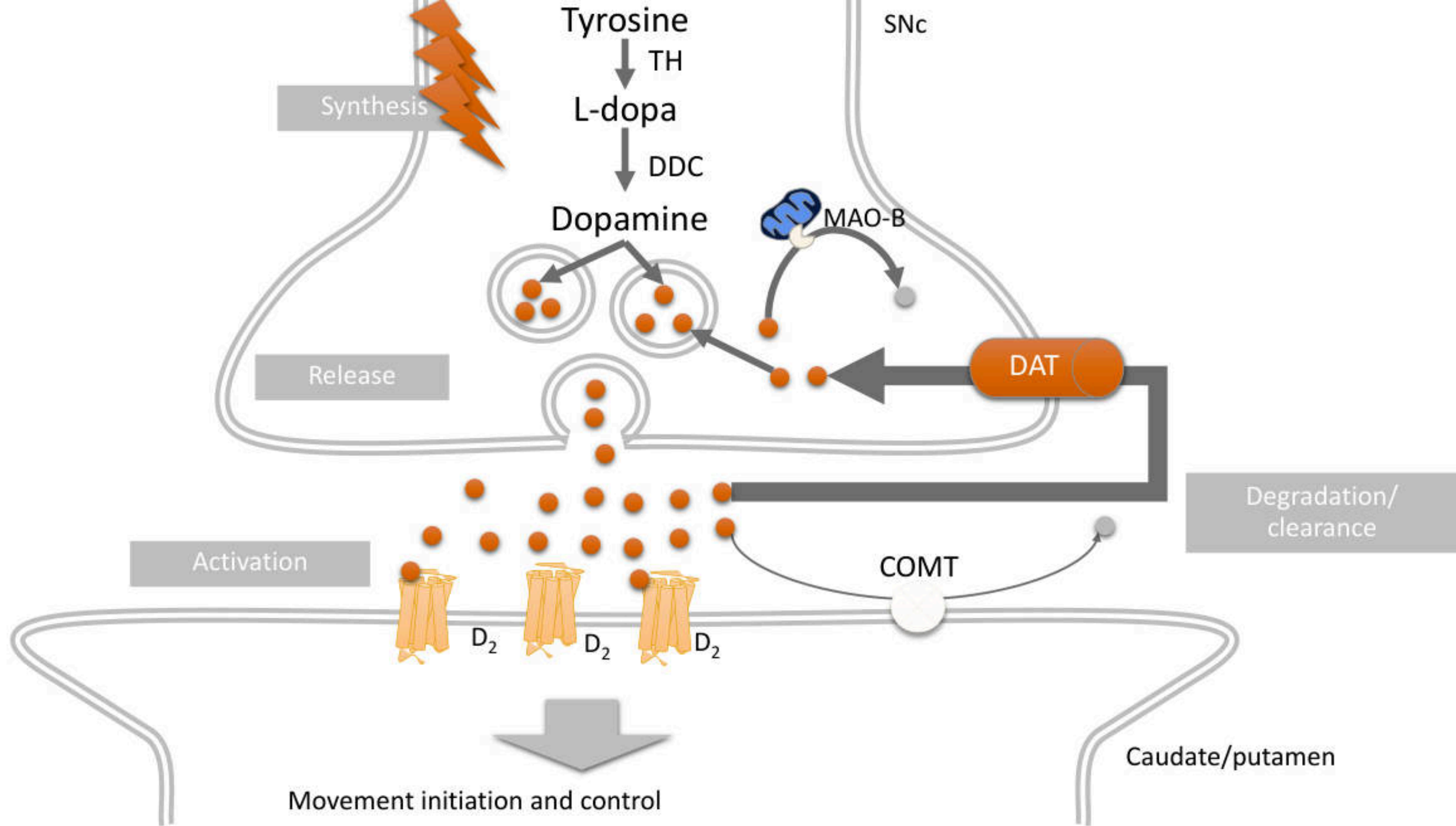


Parkinson's



Lewy bodies





Drug classes

1. Dopamine precursor: Levodopa (L-dopa) + dopa decarboxylase inhibitors
2. MAO-B inhibitors
3. COMT inhibitors
4. Dopamine receptor agonists
5. Other drugs
 - Anticholinergics
 - Amantadine

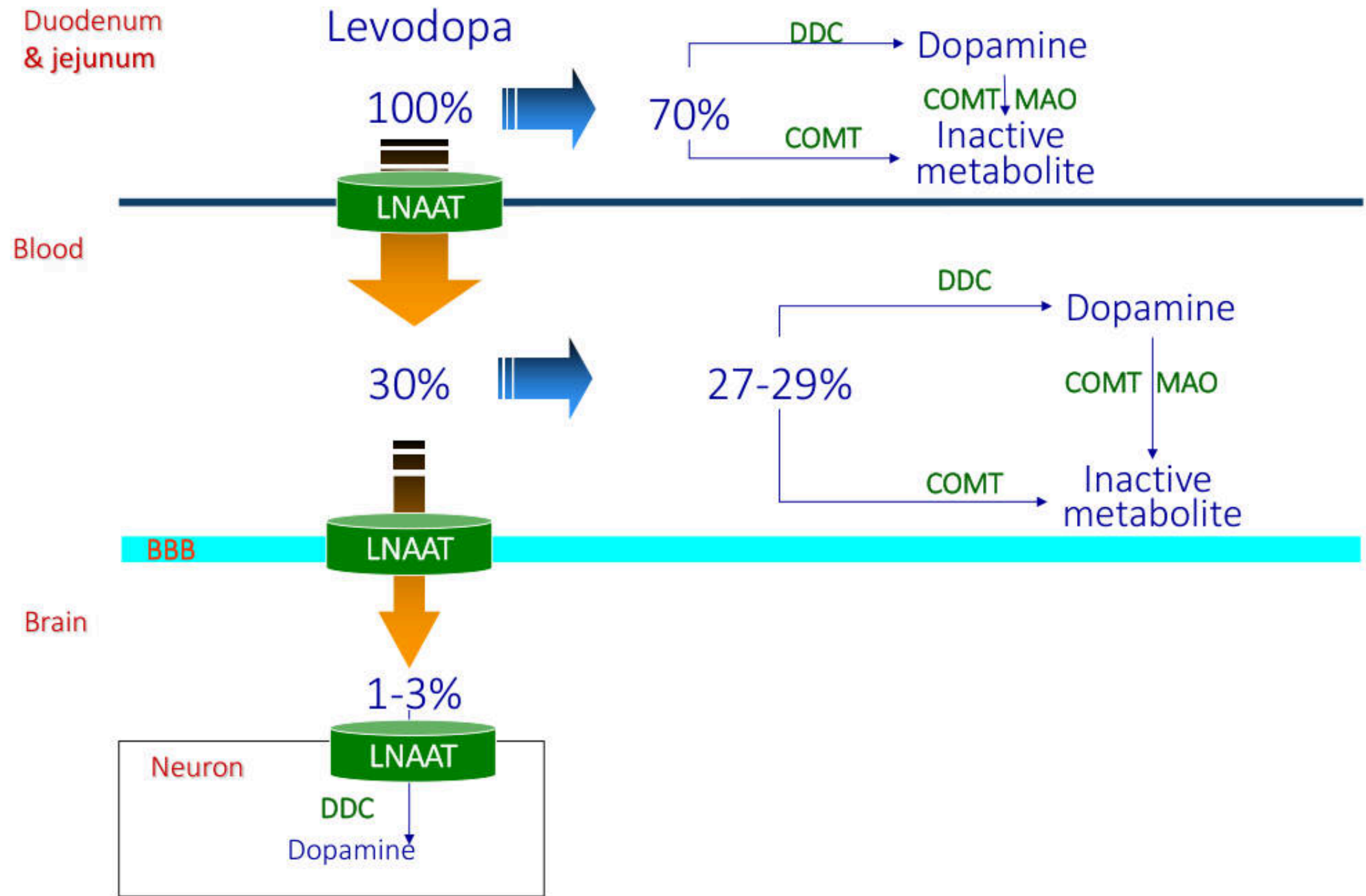
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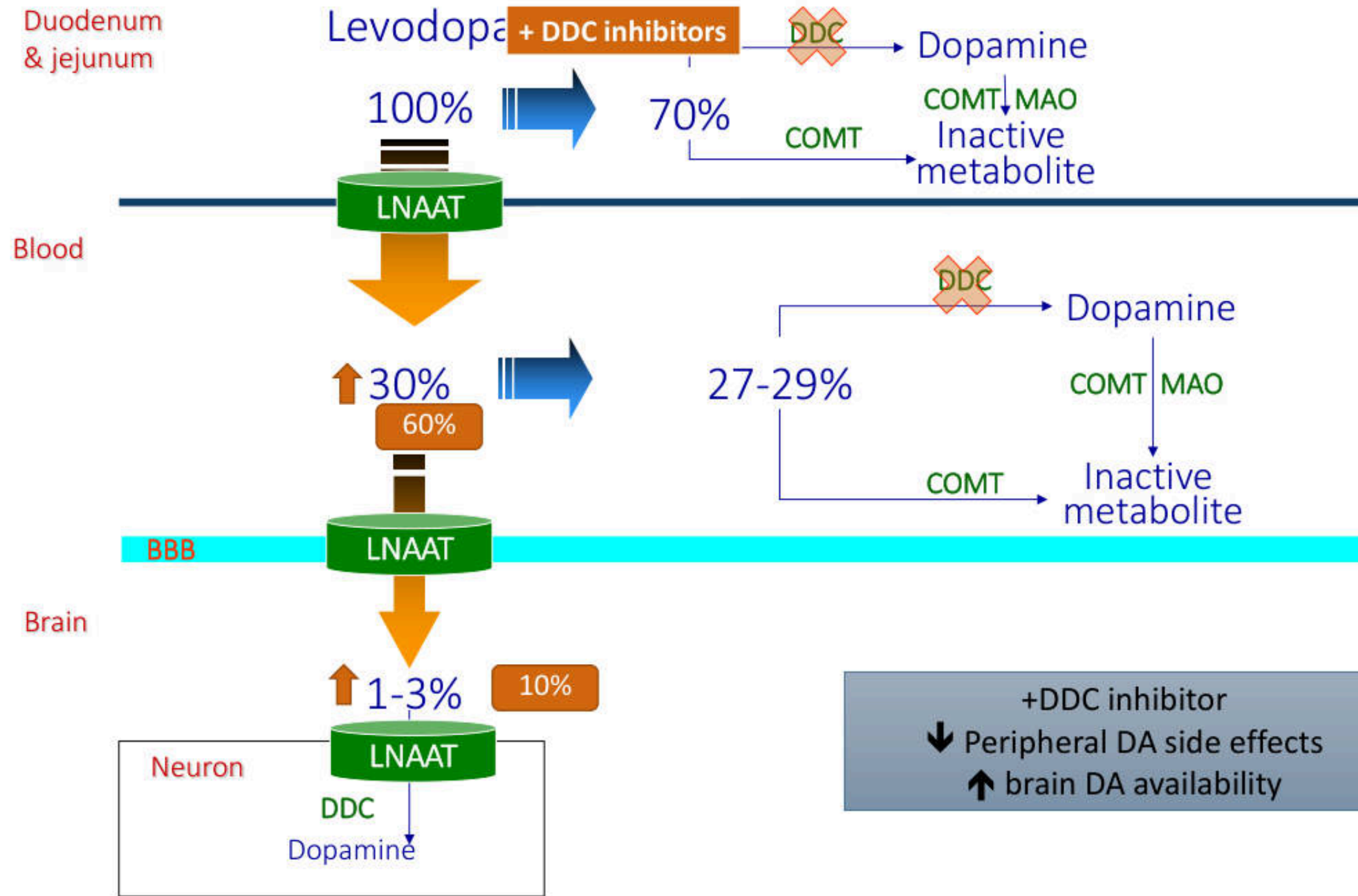
- ★ Carbidopa+levodopa
 - ◇ Sinemet®
 - ◇ Duodopa® intestinal gel
- ★ Berserazide +levodopa
 - ◇ Madopar®
 - ◇ Madopar® dispersible
 - ◇ Madopar® HBS

Gold standard for PD treatment

Levodopa

- ★ Absorbed at *small intestine* via large neutral amino acid transporter (LNAAT or LAT)
- ★ Levodopa crosses the blood-brain barrier, enters a neuron and is converted to dopamine
- ★ Levodopa is converted (by DDC) to DA in nigrostriatal neurones





Adverse effects of levodopa

- ★ Peripheral: metabolism of levodopa -> dopamine
 - ◇ CVS side effects: tachycardia, orthostatic hypotension
 - ◇ GI disturbance: nausea, vomiting, constipation

- ★ Central: increase dopamine transmission at other dopaminergic pathways
 - ◇ Mesolimbic tract
 - ◇ Tuberoinfundibular tract
 - ◇ CTZ
 - ◇ Lateral hypothalamus
 - ◇ RAS

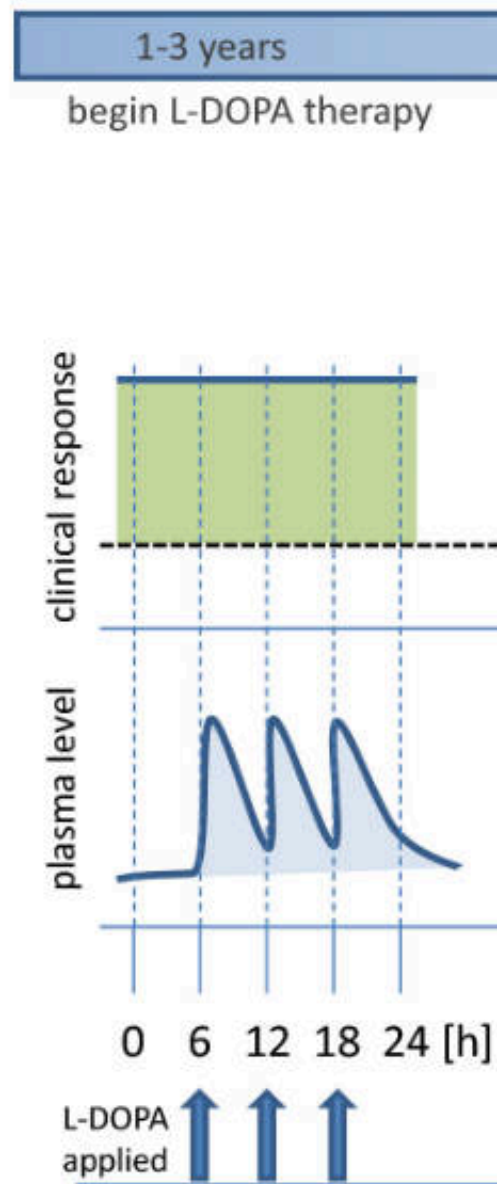
Response of levodopa in PD

★ Early PD

◇ Onset 0.5-1 hr

◇ Initial duration of action $\geq 3-6$ hr

★ Efficacy decreases over time

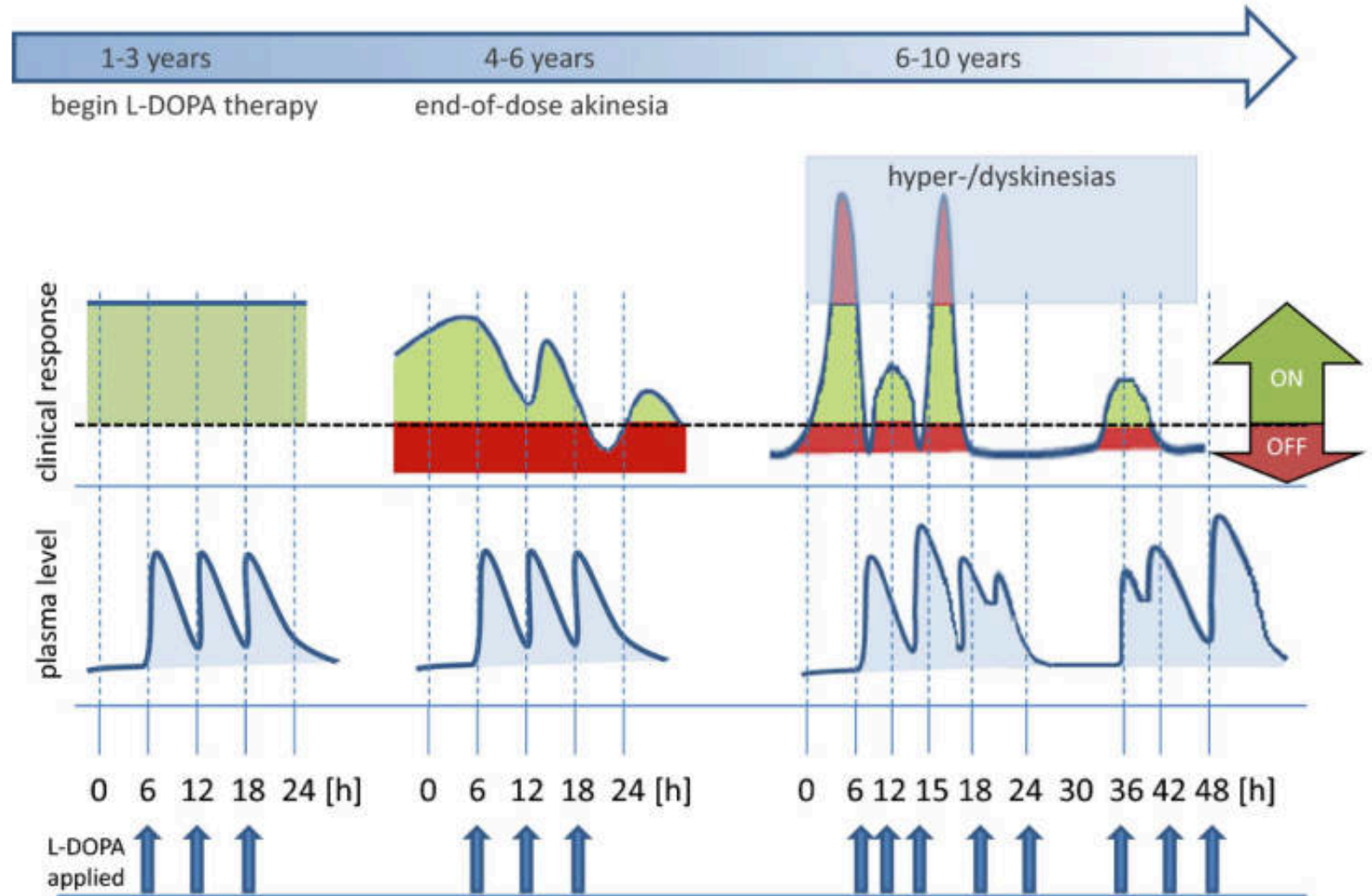


Motor fluctuation from levodopa

★ Delayed-on

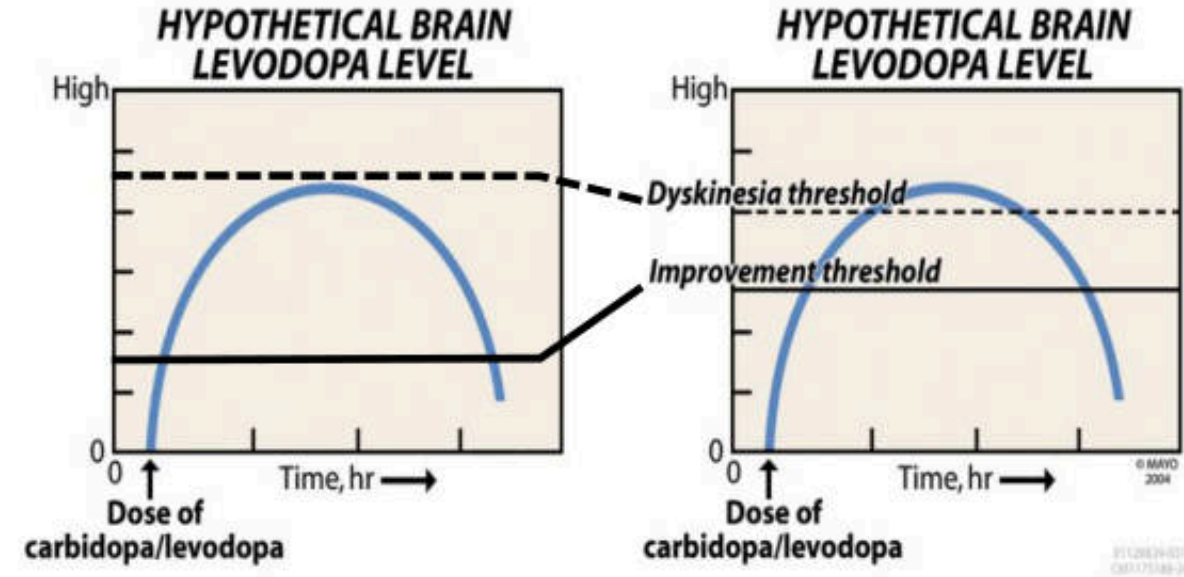
★ Long-term

- ◇ On-off phenomenon
- ◇ Wearing-off effect
- ◇ Nocturnal freezing
- ◇ Peak dose dyskinesias



Possible mechanisms of motor fluctuation from levodopa

- ★ SNC neuronal loss
 - ◇ Loss of buffering capacity
 - ◇ Narrowing therapeutic level of levodopa



- ★ Pulsatile dopamine stimulation → abnormal postsynaptic response

Treatment options for “wearing off” in patients with advanced PD

- ★ Increase dose frequency of L-dopa
- ★ Change standard formulation L-dopa to controlled-release
- ★ Add COMT inhibitor
- ★ Add dopamine agonist
- ★ Add MAOI
- ★ Add amantadine

Pharmacokinetic interactions of levodopa

- ★ Protein rich diets ↓ levodopa absorption >80%
- ★ Metal ions ↓ levodopa absorption >50%
- ★ Drugs that delay gastric emptying can delay onset of levodopa
 - ◇ Anticholinergics
 - ◇ Opioids
 - ◇ Glucagon-liked peptide-1 (GLP-1) and analogues

Patient information

- ★ Avoid taking with high protein diet
- ★ Take at least 30 min before meal or 1-2 hr after meal
- ★ Need adequate amount of fluid ideally 200 ml
- ★ Maintain upright position
- ★ Advice on nausea/vomiting and orthostatic hypotension

2. MAO-B inhibitor

- ★ Selegiline (Jumex[®], Julab[®])
- ★ Rasagiline (Azilect[®], Rasalect[®])
- ★ Safinamide (Equfina[®])

MAO-B inhibitors

- ★ Mechanism: selectively inhibit MAO-B

 - ◇ Irreversible: selegiline, rasagiline

 - ◇ Reversible: safinamide

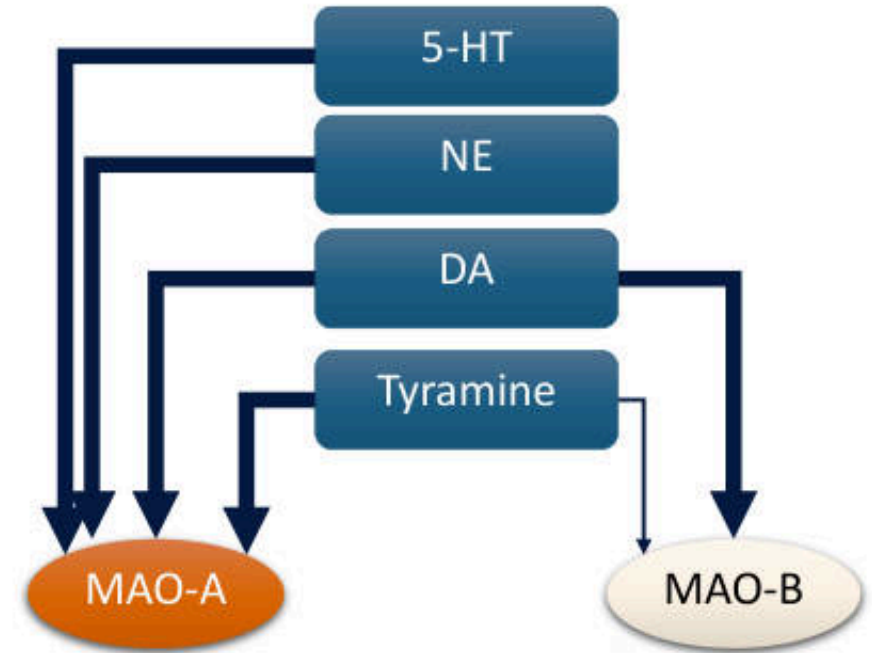
- ★ Pharmacological effects

 - ◇ Increase intracellular DA

 - ◇ Increase phasic DA release

- ★ Can be used as single agent in early but not in advanced PD

- ★ Add on to reduce motor fluctuation



Selegiline



CYP2B6/
CYP2C19



l-amphetamine

Insomnia

Take no later than 14.00

Rasagiline



CYP1A2

Aminoindan

↑ Bcl2

Disease modifying
effect

3. COMT inhibitors

- ★ Entacapone

- ◇ Entacapone alone (Comtan[®])

- ◇ Carbidopa + levodopa + entacapone (Stalevo[®])

- ★ Pharmacological effects

- ◇ ↓ peripheral conversion of levodopa to DA

- ◇ ↑ synaptic DA

- ★ Used only in combination with levodopa-> ↓ on-off phenomenon

- ★ Not for monotherapy

4. Dopamine receptor agonists

★ Ergots:

- ◆ Bromocriptine (Parlodel[®])
- ◆ Lisuride, cabergoline, pergolide

★ Non-ergots

- ◆ Pramipexole (Sifrol[®])
- ◆ Ropinirole (Requip[®])
- ◆ Rotigotine (Neupro[®] patch)
- ◆ Piribedil (Trivastal[®] Retard)
- ◆ Apomorphine (Apo-Go[®] injection)

Dopamine receptor agonists

Ergots

Bromocriptine

Mechanisms

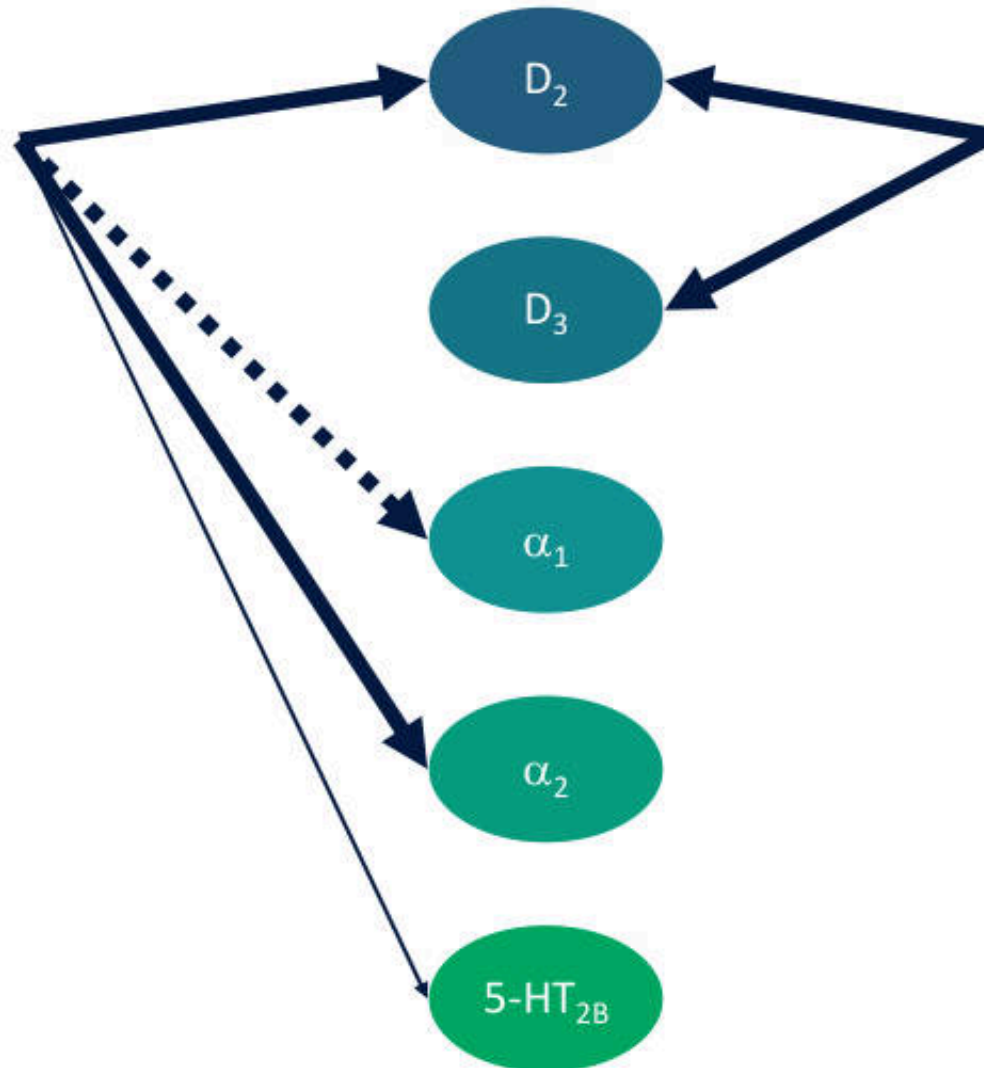
- D₂ agonist
- α_1 blockers
- α_2 agonist
- 5-HT_{2B} agonist

Non-ergots

Pramipexole,
ropinirole, rotigotine

Mechanisms

- D₂ agonist
- D₃ agonist
- Antioxidant



Adverse effects of dopamine receptor agonists

- ★ Adverse effects are due to D₂ receptor activation
- ★ Impulse control disorders e.g. gambling, punding
- ★ Ergots
 - ◇ Orthostatic hypotension, somnolence
 - ◇ Valvular fibrosis
- ★ Non-ergots: less associated with orthostatic hypotension
- ★ Sudden sleep attack
- ★ Peripheral edema (from peripheral D₂, D₃ stimulation)

Therapeutic uses of dopamine receptor agonists

★ PD

- ◇ Used in early PD → delay levodopa therapy
- ◇ Used in combination with levodopa → ↓ motor fluctuation
- ◇ Treat depression in PD especially pramipexole

★ Hyperprolactinemia

- ◇ Bromocriptine

5. Other drugs

★ Amantadine

- ◆ Dopamine transporter inhibitors and DA releaser -> Tx PD symptoms
- ◆ NMDA receptor blocker → Tx motor fluctuation
- ◆ Sigma receptor agonist → Antidepressant effect, neuroprotection
- ◆ Primarily excreted unchanged in the urine

★ Anticholinergics: trihexyphenidyl, benztropine

- ◆ Some beneficial effects in the treatment of resting tremor and muscle rigidity
- ◆ Risk > benefit
 - ✦ Anticholinergic side effect
 - ✦ Cognition
 - ✦ Delay-on